

Problem 1 - Washing Machine Mount Design

Problem 1a

The Matlab code for this problem is listed in Appendix 1.

Figure 1 shows the load force produced by the 1 kg imbalanced mass at a 0.5 m radius from the center of the washing machine as a function of angular speed in RPM.

The force amplitude is 493.5 N at a frequency of 5 Hz when the washing machine operates at 300 RPM.

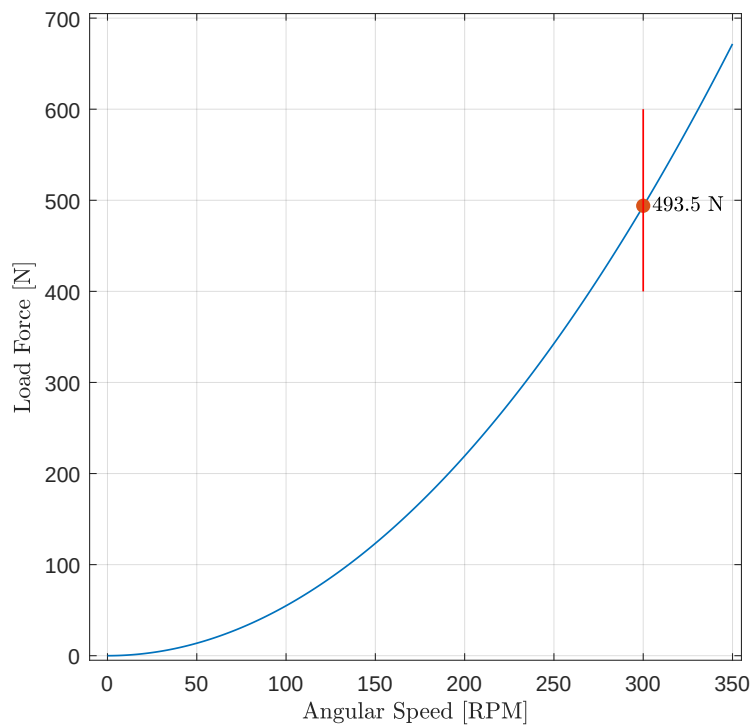


Figure 1: Load force of 1 kg imbalanced mass at a 0.5 m radius versus washing machine angular speed.

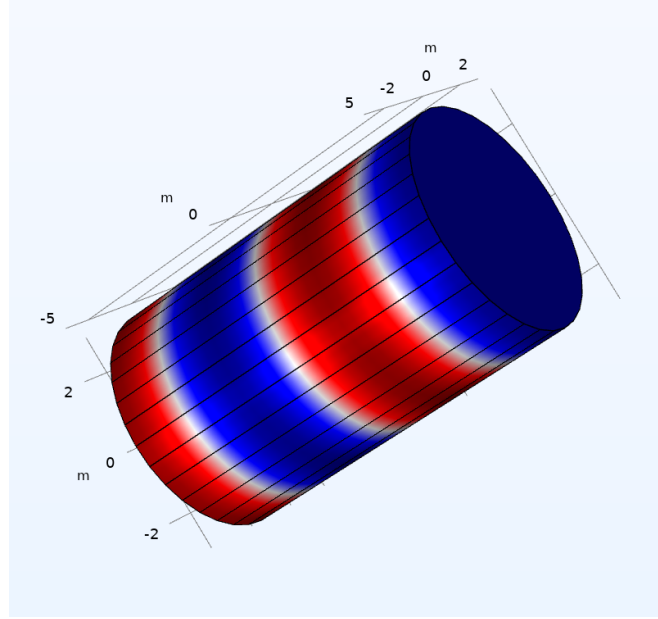
Problem 1b

The two closest modes are (3, 0, 0) and (0, 2, 0) with frequencies of 51.5 Hz and 55.6 Hz, respectively.

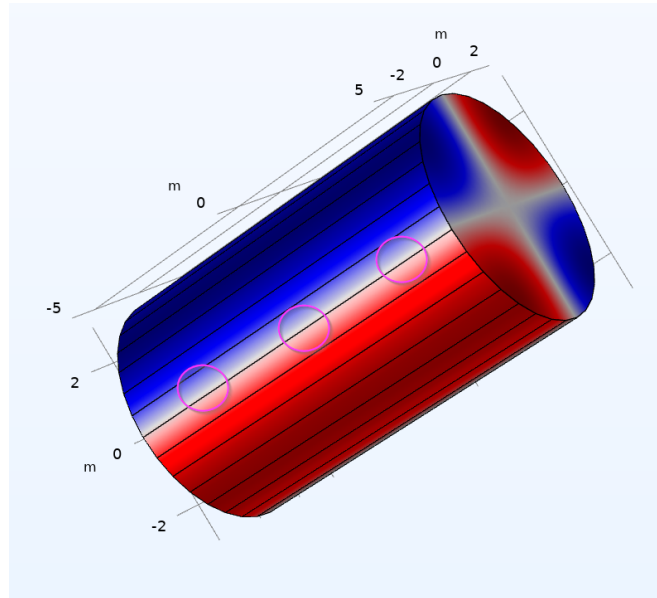
Problem 1c

Figure 2 illustrates the $(3, 0, 0)$ and $(0, 2, 0)$ modes. The white lines in each figure show the modal lines for that mode. **The machine can be placed where the modal lines for each mode overlap.** The figures were produced using the Room Eigenmode Simulator Version 1.1 software package.

The pink rings in Figure 2b indicate 3 possible places where the machine could be placed. These points coincide with the three modal planes shown in Figure 2a. Theoretically, there are an infinite number of places where the machine could be placed. However, placement would take into account practical considerations such as accessibility, etc.



(a) Mode $(3, 0, 0)$



(b) Mode $(0, 2, 0)$

Figure 2: Visualization of modes. (a.) Mode $(3, 0, 0)$. (b.) Model $(0, 2, 0)$. The pink circles in 2b illustrate a few modal line intersections where the machine could be placed.

Problem 2 - Sabine Room

The Matlab code for this problem is listed in Appendix 2.

Problem 2a

The reverberant field sound pressure level is approximately 98.3 dB SPL.

Problem 2b

Figure 3 shows the direct, reverberant, and total sound pressure levels for a 25 mW, 125 Hz, broadband, omnidirectional source placed centrally in the room (i.e. the directivity factor is 1).

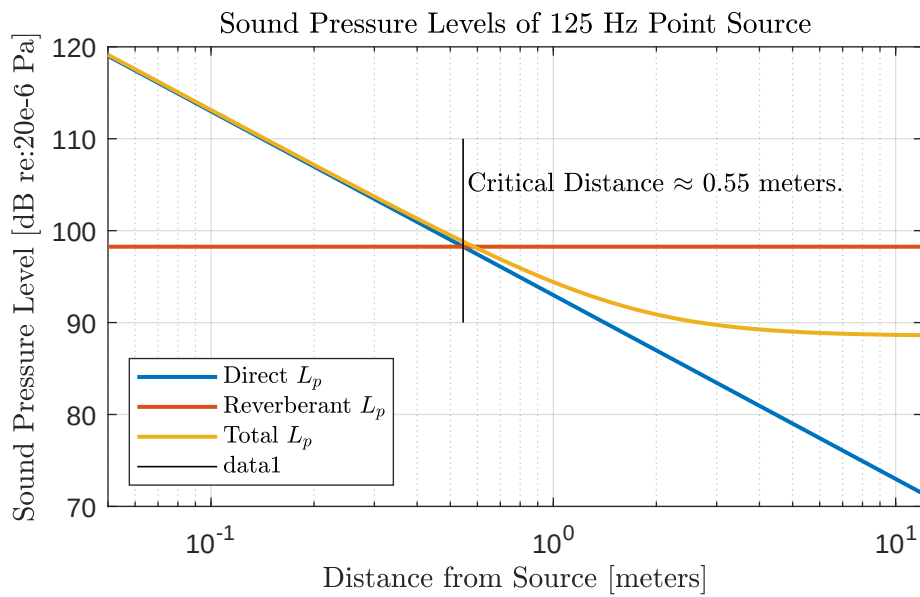


Figure 3: Sound levels for the room produced by a 25 mW, 125 Hz omnidirectional, broadband source.

Problem 3 - Transmission Loss Measurement

The Matlab code for this problem is listed in Appendix 3.

Figure 4 shows the average absorption per octave band based on the T60 data. The calibration plate isolates the receiver room and the absorption calculation does not consider the calibration plate.

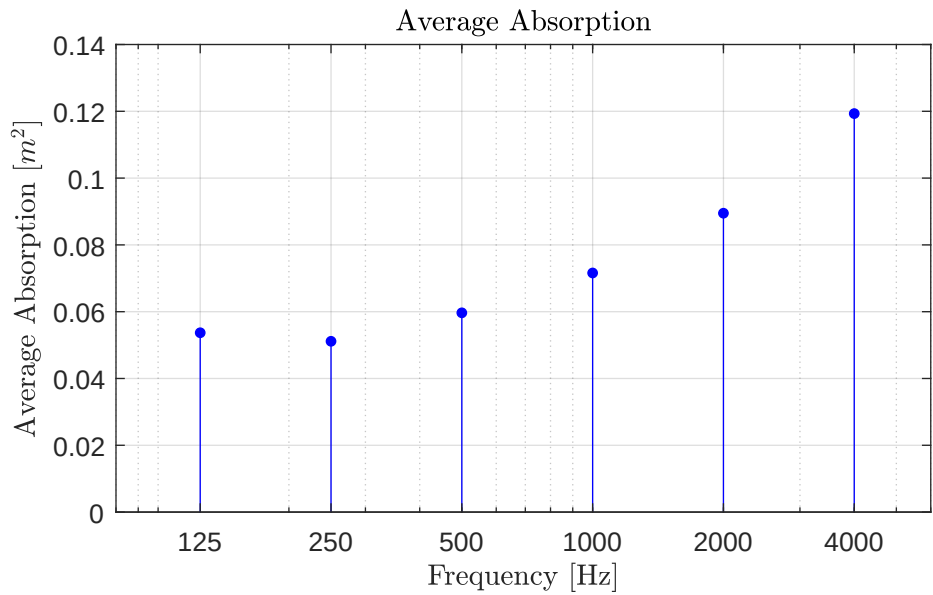


Figure 4: Average absorption per octave band.

Figure 5 shows the transmission loss per octave band.

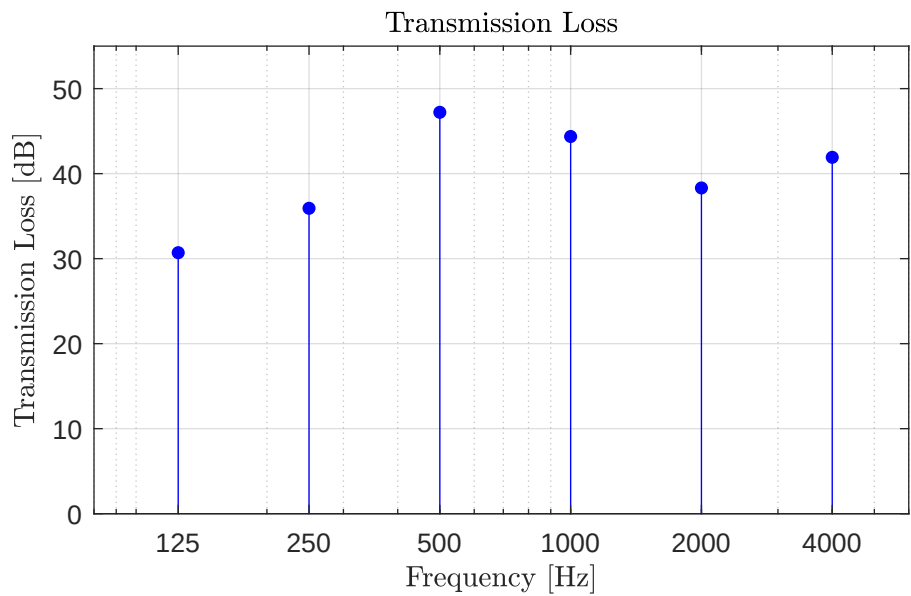


Figure 5: Transmission loss per octave band.

Problem 4 - Panel Transmission Loss

The Matlab code for this problem is listed in Appendix 4.

Problem 4a

The resonance frequency, f_0 , of the galvanized steel panel is 4 Hz or $22.6 \frac{\text{radians}}{\text{s}}$.

Problem 4b

i - Critical Frequency and Coincidence Frequency at 75°

The critical frequency is 10,216 Hz and the coincidence frequency at 75° is 10,494 Hz.

ii - Transmission Loss at Angle of Incidence of 75°

Figure 6 shows the transmission loss for an angle of incidence of 75° .

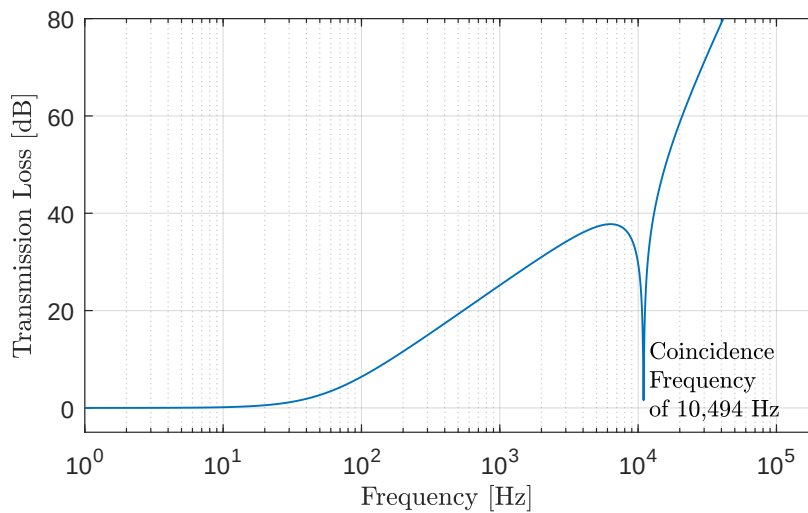


Figure 6: Flexible Panel Transmission Loss for a 75° Incidence Angle

iii - Transmission Loss for Angles of Incidence between $0-90^\circ$

Figure 7 shows the transmission loss angles of incidence from 0 to 90° in steps of 10° .

iv - Diffuse Transmission Loss

See the Matlab code in the Appendix 4

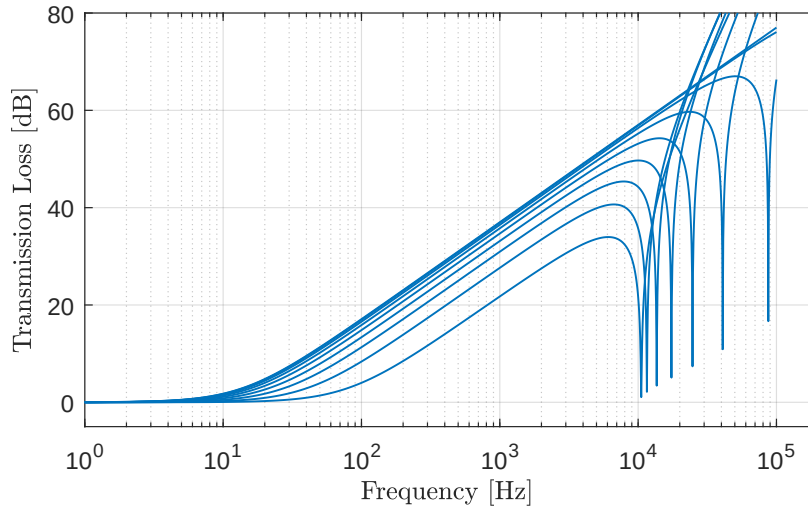


Figure 7: Flexible Panel Transmission Loss for angles of incidence between 0 and 90°.

Problem 4c

See the Matlab code in the Appendix 4

Problem 4d

Figure 8 shows the transmission loss angles of incidence from 0 to 90° in steps of 10°.

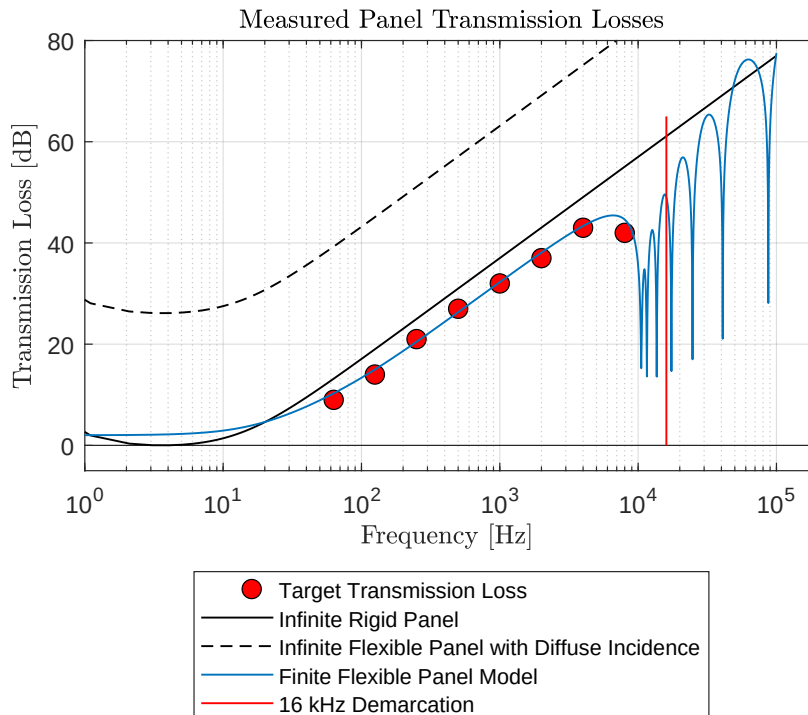


Figure 8: Measured data and the modeled responses.

The finite flexible panel appears to be the most appropriate model. As noted in class, the measured transmission loss at 8 kHz is smaller than the loss at 4 kHz. This indicates that the loss at 16 kHz should be less than the response of the infinite rigid panel at 16 kHz.

Problem 5 - Large Enclosure Design

The Matlab code for this problem is listed in Appendix 5.

My enclosure is a 2 m cube that sits on the ground (5 sides of enclosure material) over the machine. The machine is suspended above the pad (directivity factor of unity). It is assumed that there is no noise transmission through the ground.

Table 1 lists the target insertion losses, the calculated insertion losses for four materials, and the loss difference between the target and each type of material. *A positive value indicates that the target insertion loss for the given octave band was not met.*

Figure 9 shows calculated insertion loss differences for the data in Table 1.

Octave Band [Hz]	Target IL [dB]	QBV 2 [dB]	QBV 3 [dB]	HTL (100MM) [dB]	HTL 4 [dB]
250	44	28.0	19.7	10.6	5.6
500	54	26.7	22.2	6.6	-4.4
1,000	45	5.6	1.4	-15.4	-22.4
2,000	47	-8.4	-2.2	-18.4	-19.4
4,000	58	-2.4	5.7	-11.0	-13.0

Table 1: Calculated insertion losses for the 4 enclosure materials.

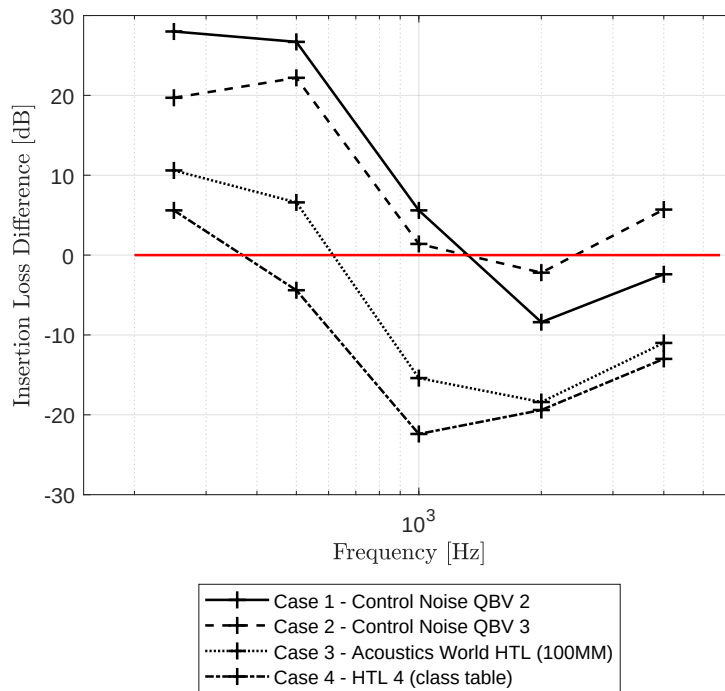


Figure 9: Insertion loss differences.

The most restrictive case given the 2 cubic-meter, 5-sided enclosure, is the HTL 4 material, Case 4, which was presented in class. The target insertion loss was reached by all of the octave bands except the 250 Hz band.

With the same enclosure dimensions and machine orientation, the HTL (100MM) material, Case 3, was found to be the second best material, not meeting the 250 Hz and 500 Hz targets. Figure 10 shows the transmission loss and absorption coefficient data for the material selected from [Acoustics World](#).

The QBV-2 and QBV-3 , Case 1 and Case 2, respectively, had the poorest performance. Figure 11 shows the transmission loss and absorption coefficient data for these material from **Control Noise**.

Using the HTL 4 material from class, a cube with 2 m sides appeared to produce the optimal insertion loss for the 250 Hz octave band. Making the size of the enclosure larger does not reduce this insertion loss and does not seem to be practical. A supplementary approach for this octave band should be considered.

NRC Rating								
Sound Absorption Coefficients (ASTM C423)								
Acoustic Panel Type	Panel Construction	125 Hz	250Hz	500Hz	1000 Hz	2000 Hz	4000 Hz	NRC
STL (100MM)	18 ga. solid / 22 ga. perforated	0.60	1.13	1.12	1.09	1.03	0.91	1.00
STL (100MM)	16 ga. solid / 22 ga. perforated	0.60	1.13	1.12	1.09	1.03	0.91	1.00
HTL (100MM)	16 ga. solid / 22 ga. perforated with HD Soundbloc Layer	0.60	1.13	1.12	1.09	1.03	0.91	1.00

STC Rating								
Sound Transmission Class								
Acoustic Panel Type	Panel Construction	125 Hz	250Hz	500Hz	1000 Hz	2000 Hz	4000 Hz	STC
STL (100MM)	18 ga. solid / 22 ga. perforated	21	28	39	48	56	58	40
STL (100MM)	16 ga. solid / 22 ga. perforated	24	32	41	51	60	66	43
HTL (100MM)	16 ga. solid / 22 ga. perforated with HD Soundbloc Layer	27	34	48	61	66	70	48

Figure 10: HTL (100MM) data from Acoustics World.

ACOUSTICAL DATA:

The **most effective** noise reduction products combine **both sound absorption and noise barrier properties**. Tested under strict compliance to appropriate ASTM standards, we offer the following results:

Sound Transmission Loss (dB) per Octave Band Frequency									
NetWell Model #	THK.	WT.	125	250	500	1000	2000	4000	STC
QBV-1	1"	1.3	11	16	24	30	35	35	27
QBV-2	2"	1.5	13	20	29	40	50	55	32
QBV-3	2.5"	2	19	25	33	46	53	58	37
QBS-1	2"	1.5	12	16	27	40	44	43	29
QBS-2	2"	2.5	19	22	28	40	56	61	33
Roof Panel	2"	2	18	24	28	37	45	46	31

Per ASTM E 90

Sound Absorption Data – Absorber Component Random Incident Sound Absorption Octave Band Center Frequencies (Hz)								
Product	THK.	125	250	500	1000	2000	4000	NRC
QBV-1	1"	.12	.47	.85	.84	.64	.62	.70
QBV-2	2"	.07	.27	.96	1.13	1.08	.99	.85
QBS-1 & 2	2"	.45	.96	.87	.66	.47	.30	.75
QB-4	4"	.21	.89	1.09	1.17	1.13	1.07	1.05

Per ASTM C 423

Flammability Ratings			
Product	Descriptor	Flame Spread	Smoke Developed
QBS-1	Vinyl faced 1" quilted fiberglass on both sides of a 1 lb. PSF non-reinforced loaded vinyl barrier septum	23	30
QBS-HT	Silicone faced 1" quilted fiberglass on both sides of a 1 lb. PSF non-reinforced noise barrier septum	4	19
QBV-2	Vinyl faced 2" quilt on one side of a 1 lb. reinforced loaded vinyl noise barrier	23	12
QBV-1	Vinyl faced 1" quilt on one side of a 1 lb. reinforced loaded vinyl noise barrier	23	30

Above table shows flame spread and smoke developed ratings per ASTM Designation E84; Surface Burning Characteristics of Building Materials. Note: Class A rating applies to products with a flame spread index of 25 or less, and a smoke developed index of 450 or less. Additional products tested to ASTM E 162 and ASTM E 662; test reports available on request.

Figure 11: QBV-2 and QBV-3 data from Control Noise.

Problem 6 - Close-fitting Enclosure Design

The Matlab code for this problem is listed in Appendix 6.

The required target insertion loss is 13 dB. This is estimated with the given data and the assumption that the sound measurement distance is beyond the critical distance (i.e., in the diffuse field).

Table 2 lists the dimensions of the designed enclosure.

Height [m]	Width [m]	Depth [m]
3	1.25	1.5

Table 2: Dimensions of the designed enclosure.

Table 3 lists the design parameters for the design enclosure.

Wall	Width [m]	Height [m]	Area [m ²]	Aspect Ratio	Correction	Compliance [$\frac{\text{m}^3 \cdot \text{s}^2}{\text{kg}}$]
Top	1.25	1.5	1.875	1.2	1.8	1.42e-6
Side 1	1.5	3	4.5	2.0	0.6	2.57e-6
Side 2	1.5	3	4.5	2.0	0.6	2.57e-6
Side 3	1.25	3	3.75	2.4	0.5	1.49e-6
Side 4	1.25	3	3.75	2.4	0.5	1.49e-6

Table 3: Enclosure design summary.

The estimated insertion loss without the hole is 13.5 dB, which exceeds the target loss of 13.

The estimated insertion loss with the hole is 1.2 dB, based on the hole depth correction of Deng (1998). The hole reduces the required insertion loss. The intake silencer needs to provide 11.8 dB of insertion loss.

1 Appendix - Matlab Code for Problem 1

2 Appendix - Matlab Code for Problem 2

3 Appendix - Matlab Code for Problem 3

4 Appendix - Matlab Code for Problem 4

5 Appendix - Matlab Code for Problem 5

6 Appendix - Matlab Code for Problem 6